Food Inspections Assignment
A Data Analyst Journey

The aim is to analyze, model, and visualize food inspection data to uncover insights and improve public health transparency

- Dealing with Food Establishment Inspections in two cities. Use below links to download data
  - Chicago and Dallas
  - Sample food inspection report of Dallas and Chicago
- All the above cities are recording data on food inspections
  - Note that both cities differ in **schema** such as The aim is to analyze, model, and visualize food inspection data to uncover insights and improve public health transparency
- **Tools to be used:**
  - Databricks for processing, data profiling and cleansing via DQX (If you want you can use ADF)
  - ER studio / Navicat for data modeling

Food Establishment Inspections

Each part 25 Marks

- Deliverables – follow Medallion architecture wherever applicable
  - Part 1
    - Profile the data and document the findings (DQX)
      - Goal is to understand structure, quality, characteristics
      - Helps to find common attributes to create a dataset which can be merged and loaded to target.
  - Part 2
    - Design Dimensional Model
    - Source to target mapping document to be created.
    - Deploy the model (Databricks final tables should match to this model)
  - Part 3
    - Load dimensional model
      - Read silver zone (cleansed) and load the model (Gold zone)
      - Name Dimensions and Facts with appropriate prefix such as dim_ / fact_
      - Implement SCD2 for any one of your choice of dimension table.(You can either use DLT or Manual implementation)
  - Part 4
    - Create BI dashboards in any one of the BI tool
      - Power BI / Tableau

Food Establishment Inspections

- Silver zone data validation rules to be incorporated as expectations (drop the bad rows)
  - Restaurant Name, Inspection Date and Type, Zip codes cannot be null
  - Zip codes cannot be null and should be in valid format
  - Violation score in Dallas cannot be more than 100
  - Chicago inspection results cannot be null
  - Every inspection must have at least 1 unique violation and duplication violations to be loaded as distinct
  - In Dallas dataset –
    - if violation score is 90 or more than we should not have more than 3 violation associated
    - Inspection result cannot be PASS if any of the violation contains Urgent/Critical terms
- Derive violation score for Chicago based on the table provided here →
- Remember –
  - All good practices discussed in the class must be incorporated
  - Profiling and analysis to be clearly documented
  - Cleansing Steps and adjustments performed to be documented
  - Submit screen shots clearly showing the end results
  - Code must be integrated to GIT as explained in class
  - Will grab the source code from GIT for grading
  - Git commits post due date will not be considered
  - Note that violation codes and descriptions in Chicago are not structured
    - Standardize the schema for violation codes and descriptions between both datasets so that you can store data properly
    - Dallas violations codes and Chicago violation codes dont need to match however, create a dim that stores both sets

Chicago Results	Derive score based on Results
Pass	90
Pass w/ Conditions	80
Fail	70
No Entry	0
All other types	NULL

What to show on Dashboard

Expected Visuals

- Examine food inspection results by:
  - Inspection result
  - Inspection type
  - Risk category
  - Facility type
  - Violations (Codes, descriptions)
  - Business inspected
    - DBA (Doing Business As), AKA (Also Know As), License
  - Location
- Inspection report
  - All above with inspection #, license #, violations & inspector comments